Dipartimento di Elettronica, Informazione e Bioingegneria

Politecnico di Milano

Prof. Matteo Camilli, Elisabetta Di Nitto, and Matteo Rossi

20133 Milano (Italia)
Piazza Leonardo da Vinci, 32
Tel. (39) 02-2399.3400
Fax (39) 02-2399.3411

Software Engineering 2 – Pre-exam

December 19th, 2024

Notes

A. This file contains the description of the problem that you are asked to model in Alloy. The solution must be provided through the following Microsoft form: https://forms.office.com/e/apgRF9hyJ2

B. You must use the Alloy Analyzer (version 6.1.0) to create the file with your proposed solution.

C. Follow the instructions provided in the form to submit your Alloy model (which you need to provide through various text boxes).

D. **If the signatures, combined with the facts and predicates of Part 1, do not compile and do not produce reasonable instances, the rest of the exam (Part 2 and Part 3) will not be evaluated.**

E. **Total available time: 2h and 30 mins**

F. **Remember that:**

1) Non-compiling signatures and/or non-compiling facts and predicates of Part 1 will result in zero points, also disqualifying students from the first exam call (January 23, 2025).
2) An overall score lower than 5 points (out of 16) will disqualify students from the first exam call (January 23, 2025).

G. **Organization of the submission form**: The exam form includes one text box dedicated to the signatures and integrity facts you will create and modify while addressing the three parts of the exam.

While addressing parts 2 and 3 you may need to modify the signatures/integrity fact defined in the previous parts. In this case, please update your code in the signatures and facts box as needed and make sure that all predicates and assertions in the previous parts still work properly.

H. **Assessment criteria.** Our assessment will be based on the following criteria:

1) The extent to which your model and associated comments address the specific requests in the exam.
2) The coherence of your model with the specific domain, which, in this case concerns the Kafka system, in particular, partitions, partition replicas, and broker leadership.

Question – Alloy (16 points)

Consider the elements and features of a Kafka application. This is composed of the following elements: one or more message producers, some Kafka clusters, and one or more message consumers (see Fig. 1). A Kafka cluster is made of brokers, and each broker hosts messages. Each message belongs to a topic. The messages of each topic are organized in partitions. Each broker can host one or more partitions of a topic. Also, partitions can be replicated across brokers of the same cluster. Each partition has a leading broker and can have zero or more follower brokers. The leading broker of a partition is the one that receives from message producers the messages associated with the corresponding partition and distributes them to the follower brokers (if any). Consumers can contact either the leading broker or the follower brokers to receive the messages associated with the partition (this point is included to clarify how consumers use the Kafka cluster but should not be modeled in Alloy).
Brokers can fail. In this case, they (and the partitions they host) disappear from the cluster.

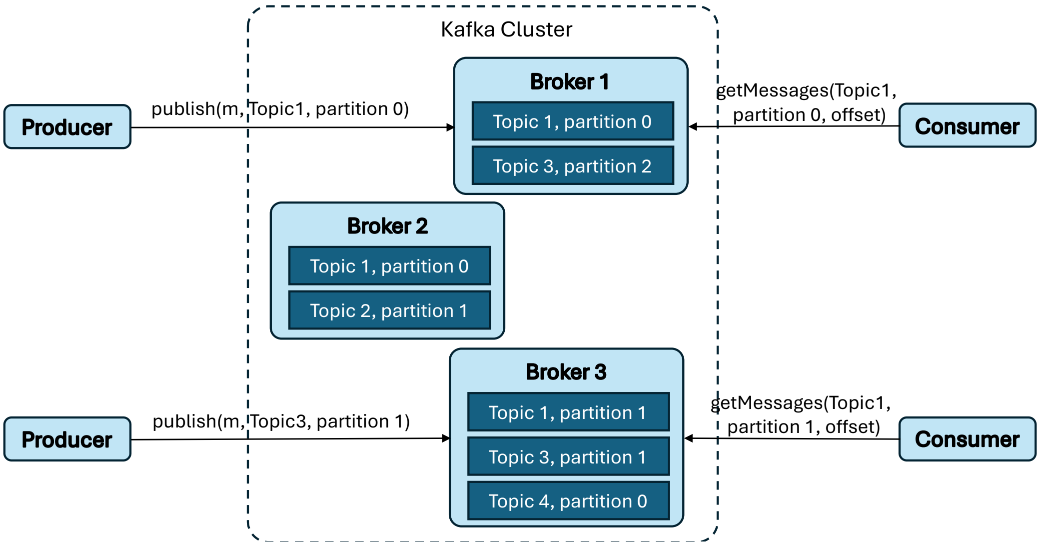

Figure 1. Kafka application example.

Part 1

1. Define suitable Alloy signatures (and related constraints) that capture Kafka clusters. In this part, you shall not deal with message reception or brokers failing (they are the topic of parts 2 and 3).
2. Define and execute a predicate that shows an instance of a single Kafka cluster that includes at least 2 brokers, each with at least 2 partitions of different topics. Also, there must be at least one partition that is replicated across some of the brokers.

Part 2

When a message is delivered to the leading broker of a partition, the latter distributes the message to each follower broker. The distribution occurs instantaneously, in the next instant after the message is received by the leading broker. For simplicity, we consider that messages are never deleted from partitions, they are only added.

1. Introduce suitable elements (e.g., constraints) that define that when a message is delivered to the Kafka cluster, if it is delivered to a replicated partition, it is first stored in the partition of the leading broker; then, at the next step, it is spread to the partitions on the follower brokers.
2. Define a predicate that shows an evolution of a Kafka cluster in which 2 messages, concerning different topics, are delivered to the leading broker of a replicated partition over 2 consecutive time instants. Execute the predicate to show that the behavior is feasible in your model.
   In this point, you can neglect the issue that brokers can fail and consider that they never disappear from the cluster.

Part 3

We now want to represent the possibility that some brokers fail, that is, they disappear from the Kafka cluster, and a new leading broker is selected for the replicated partitions for which they are leaders.

1. Introduce suitable elements in the Alloy model that allow a broker to fail (that is, it simply disappears from the cluster) and the partitions running on it to disappear from the cluster. In case the broker is leader of some replicated partition, a new leader is selected among the followers of the replicated partition. You shall not explicitly model the policy to choose the new leader.
2. Define and execute a predicate that produces an instance of the model in which a broker that is the leader of a replicated partition fails.
3. Define an assertion to check whether a partition will always have the same leader. Check the assertion and explain the result through a comment in the model.